CANCER TREATMENT APPLICATION BASED ON PUBMED ARTICLES

Medical AI Chatbot: Intelligent Healthcare Assistance Using Deep Learning

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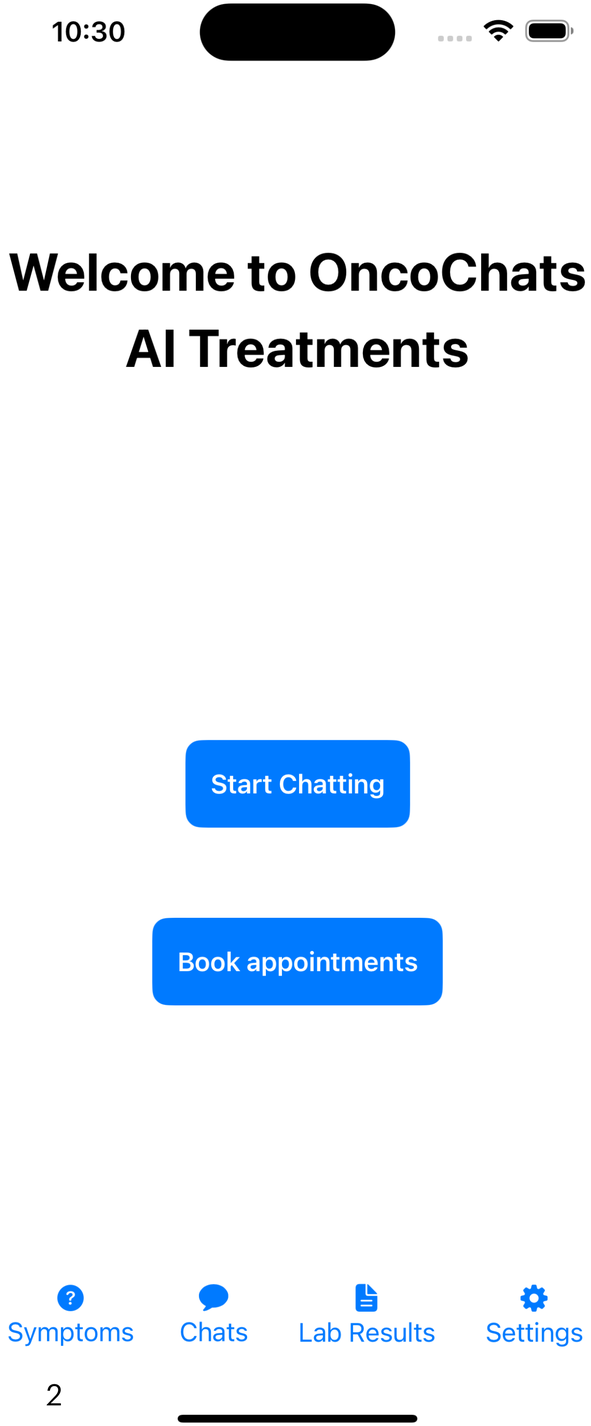
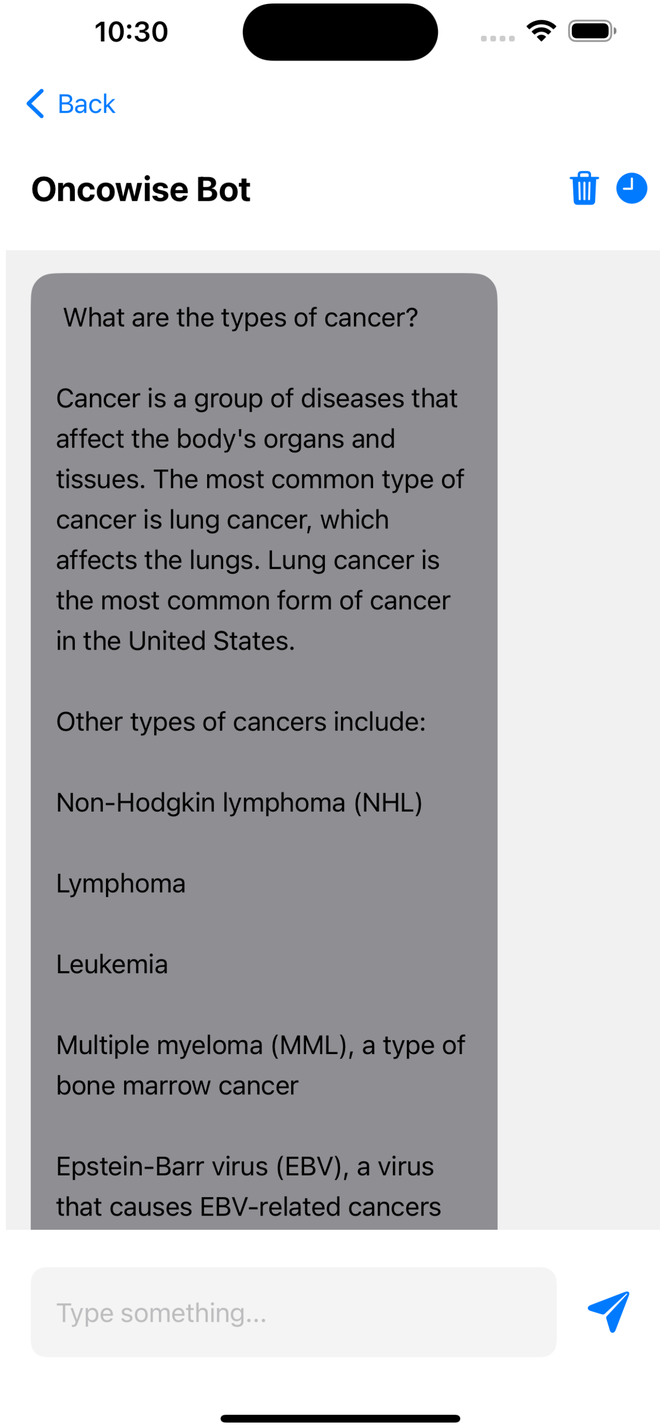




Problem Statement

1.1 Current Challenges in Medical Information Access

- **Information Overload:** Medical professionals and patients face challenges in quickly accessing relevant medical information
- **Complex Medical Terminology:** Existing information retrieval systems struggle with domain-specific language and nuanced medical contexts
- Need for Intelligent Information Processing:

➡Rapid and accurate medical query resolution

➡Contextually aware response generation

➡Bridging the gap between medical knowledge bases and user-friendly communication

Problem Statement

1.2 Research Objectives

Develop an AI-powered medical chatbot

Key Innovation Areas:

- Precise medical query classification
- Accurate knowledge extraction
- Contextually refined response generation

Targeted Impact:

- Assist medical professionals in quick information retrieval
- Provide accessible medical insights for patients

Methodology

2.1 Integrated Architectural Approach

- **Classification Module:** Uses **BioBERT fine-tuned** to categorize user queries (e.g., symptoms, diseases, treatment).
- **Question Answering Module:** Employs **BioBERT fine-tuned on MedQuAD** to extract medical knowledge from given contexts.
- **Response Refinement Module:** A **fine-tuned GPT-2 model** rephrases AI-generated responses to ensure natural language flow.
- **Backend:** Flask-based API deployed for accessibility.
- **Client Side:** A Swift-based application that interacts with the Flask server for real-time responses.

3. System Architecture

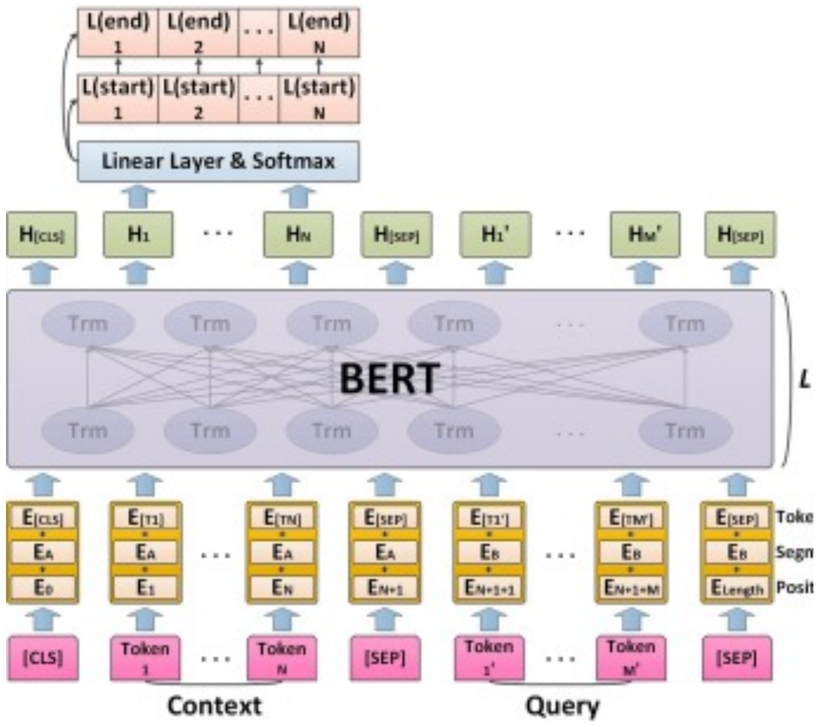
User Input Processing

1. Swift-based mobile application interface
2. Secure query transmission via Flask API
3. Real-time communication protocol

Backend Intelligence Pipeline

1. Classification Layer:
    - BioBERT identifies query category
    - Routing to appropriate processing module
2. Knowledge Extraction Layer:
    - BioBERT-QA retrieves relevant medical information
    - Contextual understanding of query intent
3. Response Generation Layer:
    - Fine-tuned GPT-2 refines extracted information
    - Ensures natural, readable medical explanations

3. System Architecture



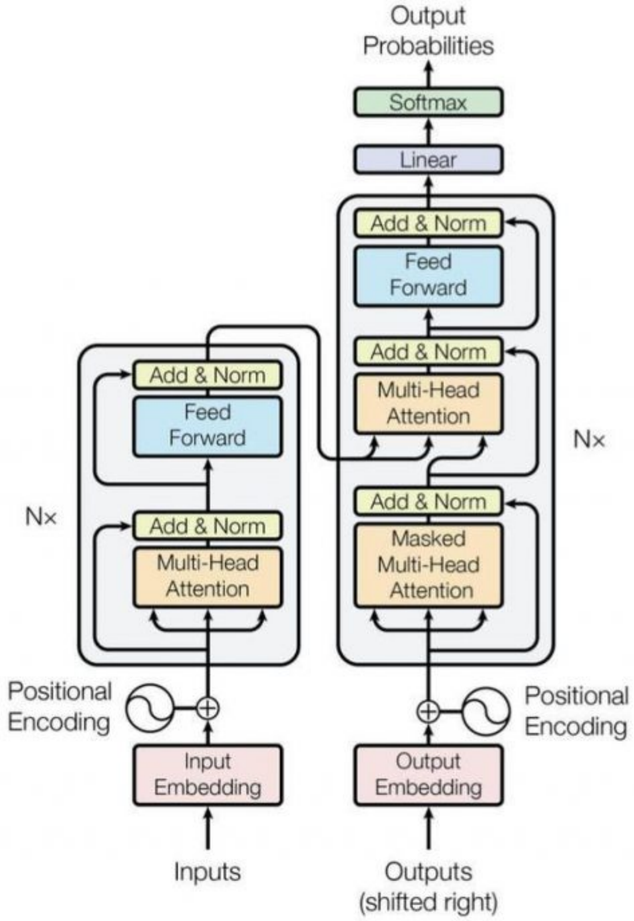


Figure 1: The Transformer - model architecture.

BioBERT Classification model

Text generative model

4. Implementation Details

4.1 Dataset Composition

For Question and answer model:

MedQuAD Dataset from PUBmed article

For classification model:

All cancer types model from Kaggle

For gpt model:

Health Counseling Conversations Dataset from Kaggle

4.2 Dataset information for Q&A model

MedQuAD Dataset:

- Comprehensive medical question-answer pairs
- Diverse medical domain coverage
- Ground truth for model training and validation
- 16407 QA pairs from MedQuAD dataset

4. Implementation Details

4.2Dataset Raw format for Q&A model


```
<QAPairs>
        <QAPair pid="1">
                        <Question qid="0000001_1-1" qtype="information">What is (are) Adult Acute Lymphoblastic Leukemia ?</Question>
                        <Answer>Key Points
                    - Adult acute lymphoblastic leukemia (ALL) is a type of cancer in which the bone marrow makes too many lymphocytes (a type of white b
Leukemia may affect red blood cells, white blood cells, and platelets.    - Previous chemotherapy and exposure to radiation may increase the risk of deve
Signs and symptoms of adult ALL include fever, feeling tired, and easy bruising or bleeding.     - Tests that examine the blood and bone marrow are used
diagnose adult ALL.    - Certain factors affect prognosis (chance of recovery) and treatment options.


                    Adult acute lymphoblastic leukemia (ALL) is a type of cancer in which the bone marrow makes too many lymphocytes (a type of white blo
                    Adult acute lymphoblastic leukemia (ALL; also called acute lymphocytic leukemia) is a cancer of the blood and bone marrow. This type
gets worse quickly if it is not treated.


                    Leukemia may affect red blood cells, white blood cells, and platelets.
                    Normally, the bone marrow makes blood stem cells (immature cells) that become mature blood cells over time. A blood stem cell may bec
cell or a lymphoid stem cell.   A myeloid stem cell becomes one of three types of mature blood cells:          -  Red blood cells that carry oxygen and o
all tissues of the body.    -  Platelets that form blood clots to stop bleeding.    -  Granulocytes (white blood cells) that fight infection and disease.
stem cell becomes a lymphoblast cell and then one of three types of lymphocytes (white blood cells):         -  B lymphocytes that make antibodies to hel
-  T lymphocytes that help B lymphocytes make the antibodies that help fight infection.    -  Natural killer cells that attack cancer cells and viruses.
many stem cells become lymphoblasts, B lymphocytes, or T lymphocytes. These cells are also called leukemia cells. These leukemia cells are not able to fi
well. Also, as the number of leukemia cells increases in the blood and bone marrow, there is less room for healthy white blood cells, red blood cells, an
cause infection, anemia, and easy bleeding. The cancer can also spread to the central nervous system (brain and spinal cord).     This summary is about a
lymphoblastic leukemia. See the following PDQ summaries for information about other types of leukemia:          -  Childhood Acute Lymphoblastic Leukemia
Adult Acute Myeloid Leukemia Treatment.    -  Childhood Acute Myeloid Leukemia/Other Myeloid Malignancies Treatment.    -  Chronic Lymphocytic Leukemia T
Chronic Myelogenous Leukemia Treatment.    -  Hairy Cell Leukemia Treatment.</Answer>
        </QAPair>
```

4. Implementation Details

4.2Preprocessing Dataset format for Q&A model

- **Tokenization**

Splitting the question and answer into individual words or subwords using a tokenizer (e.g., WordPiece for BioBERT).

- **Adding Special Tokens**

Adding [CLS] at the beginning of the input and [SEP] between the question and answer to mark separation.

[CLS] How many people are affected by cell hypoplasia ? [SEP] Cell hypoplasia is a rare disorder ; its prevalence is unknown . [SEP]

- **Padding and Truncation**

Ensuring that all inputs have the same length by padding shorter sequences with special tokens (like [PAD]) or truncating longer ones.

- **Converting to Token IDs**

Mapping the tokens to their corresponding IDs in the BioBERT vocabulary.

- **Creating Attention Masks**

Generating a binary mask where 1 indicates a real token and 0 represents padding.

- **Formatting for Model Input**


```
},
{
    "question": "How many people are affected by Leydig cell hypoplasia ?",
    "answer": "Leydig cell hypoplasia is a rare disorder; its prevalence is unknown."
},
{
```

4. Implementation Details

4.2Dataset processing for classification model

1. Preprocess Data

- Clean text (remove unwanted characters, fix encoding issues).
- Tokenize using **BioBERT tokenizer**.
- Convert text into input IDs, attention masks.
- Encode labels (convert text labels like "Thyroid_Cancer" into numerical values).

2. Model Selection

- Use BioBERT with a classification head (a linear layer on top of [CLS] token).
- Loss function: **Cross-Entropy Loss** for multi-class classification.

	Types	Reports
0	Thyroid_Cancer	Thyroid surgery in children in a single institution from Osama Ibrahim Almosallama Ali Aseerib Ahmed Alhumaida Ali S AlZahranic Saif Alsobhib remaining cases underwent FNA at another institution and FNA was not repeated at our institution or they came for completion thyroidecton McLaren GD Nichaman MZ Iodine and goiter in children Pediatrics Ries LAG Melbert D Krapcho M Stinchcomb DG Howlader N Horner MJ
1	Thyroid_Cancer	" The adopted strategy was the same as that used in prior years [] and is based on four exclusive queries that return four disjoint citation sub
2	Thyroid_Cancer	coronary arterybypass grafting thrombosis п¬Ґbrin п¬Ґbrinogen mutationIntroduction Intraoperative thrombosis of saphenous veins SV during

4. Implementation Details

4.2Dataset processing for gpt2

1. Preprocess Data

- Clean text (remove unwanted characters, fix encoding issues).
- Tokenize using **GPT-2 tokenizer**.
- Convert text into input IDs, attention masks.
- Encode labels .
- Remove special characters & unnecessary symbols
- Lowercase conversion (if needed)
- Merge conversations into a structured format


```
"Context" :
string "I'm going through some things with my feelings and myself. I barely sleep and I do nothing but
think about how I'm worthless and how I shouldn't be here. I've never tried or contemplated suicide. I'v
always wanted to fix my issues, but I never get around to it. How can I change my feeling of being
worthless to everyone?"
"Response" :
string "If everyone thinks you're worthless, then maybe you need to find new people to hang out
with.Seriously, the social context in which a person lives is a big influence in self-esteem.Otherwise,
you can go round and round trying to understand why you're not worthless, then go back to the same crowd
and be knocked down again.There are many inspirational messages you can find in social media.  Maybe rea
some of the ones which state that no person is worthless, and that everyone has a good purpose to their
life.Also, since our culture is so saturated with the belief that if someone doesn't feel good about
themselves that this is somehow terrible.Bad feelings are part of living.  They are the motivation to
remove ourselves from situations and relationships which do us more harm than good.Bad feelings do feel
terrible.   Your feeling of worthlessness may be good in the sense of motivating you to find out that yo
are much better than your feelings today."
}
```

4. Implementation Details

4.3 Model Configuration

BioBERT Model:

- **Pre-training:** Biomedical literature corpus
- Fine-tuning Approach:
  - Task-specific adaptation
  - Multi-label classification capabilities
  - Contextual embedding refinement

GPT-2 Model:

- Generative Pre-training:
  - Natural language understanding
  - Coherent text generation
- Domain-Specific Adaptation:
  - Medical terminology integration
  - Conversational response generation

4. Implementation Details

4.4 Optimization Strategies

4.4 Optimization Strategies

- Training Techniques:
  - **AdamW:** Adaptive learning
  - With a small learning rate (default:2e-5) for stability

Why AdamW?

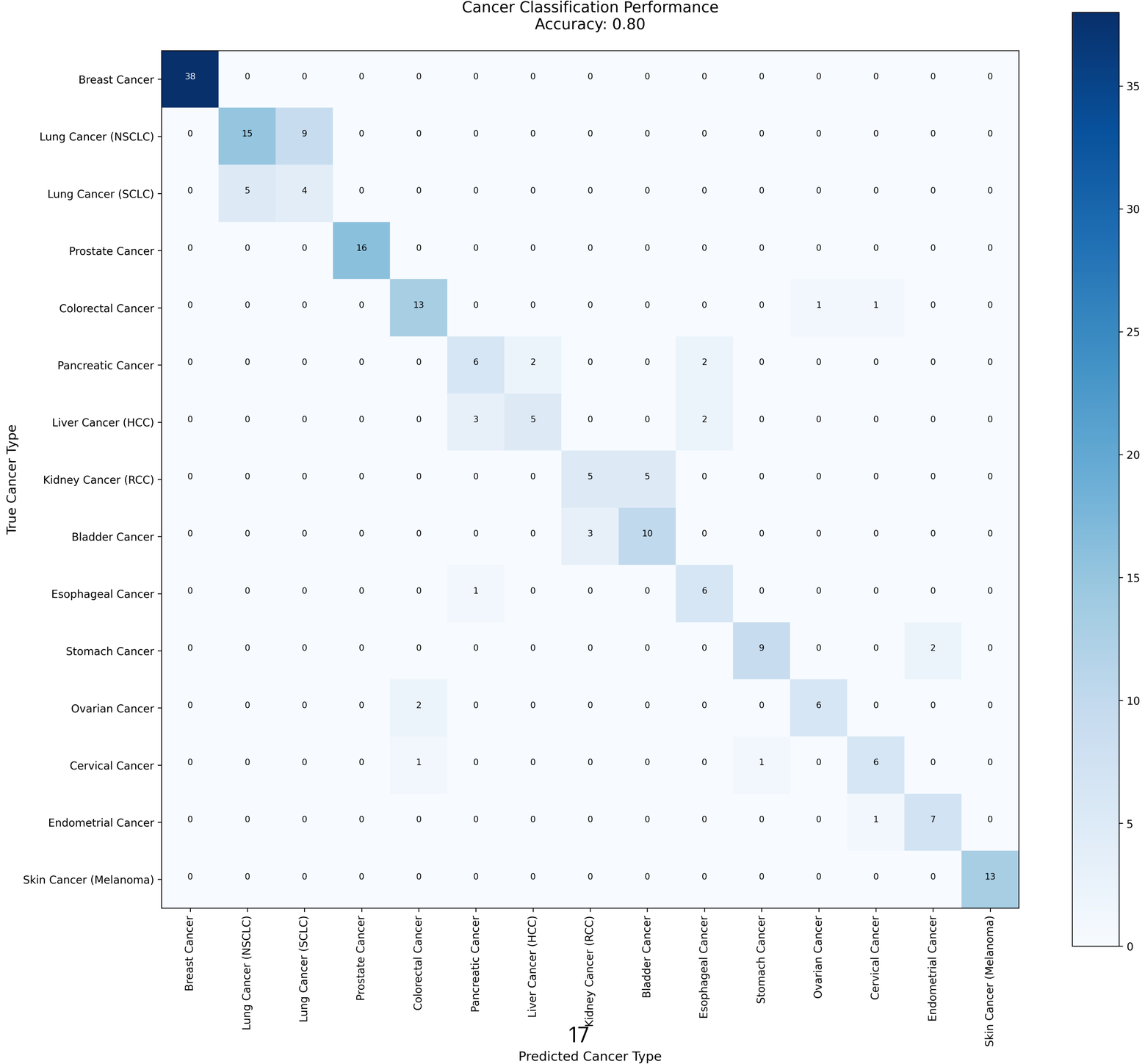
- Prevents overfitting with weight decay
- Works well with transformer-based models

5. Result

5.1 Performance Metrics for Classification

Classification Accuracy:

- Precise medical query categorization
- Minimal misclassification rates
- High-confidence query routing

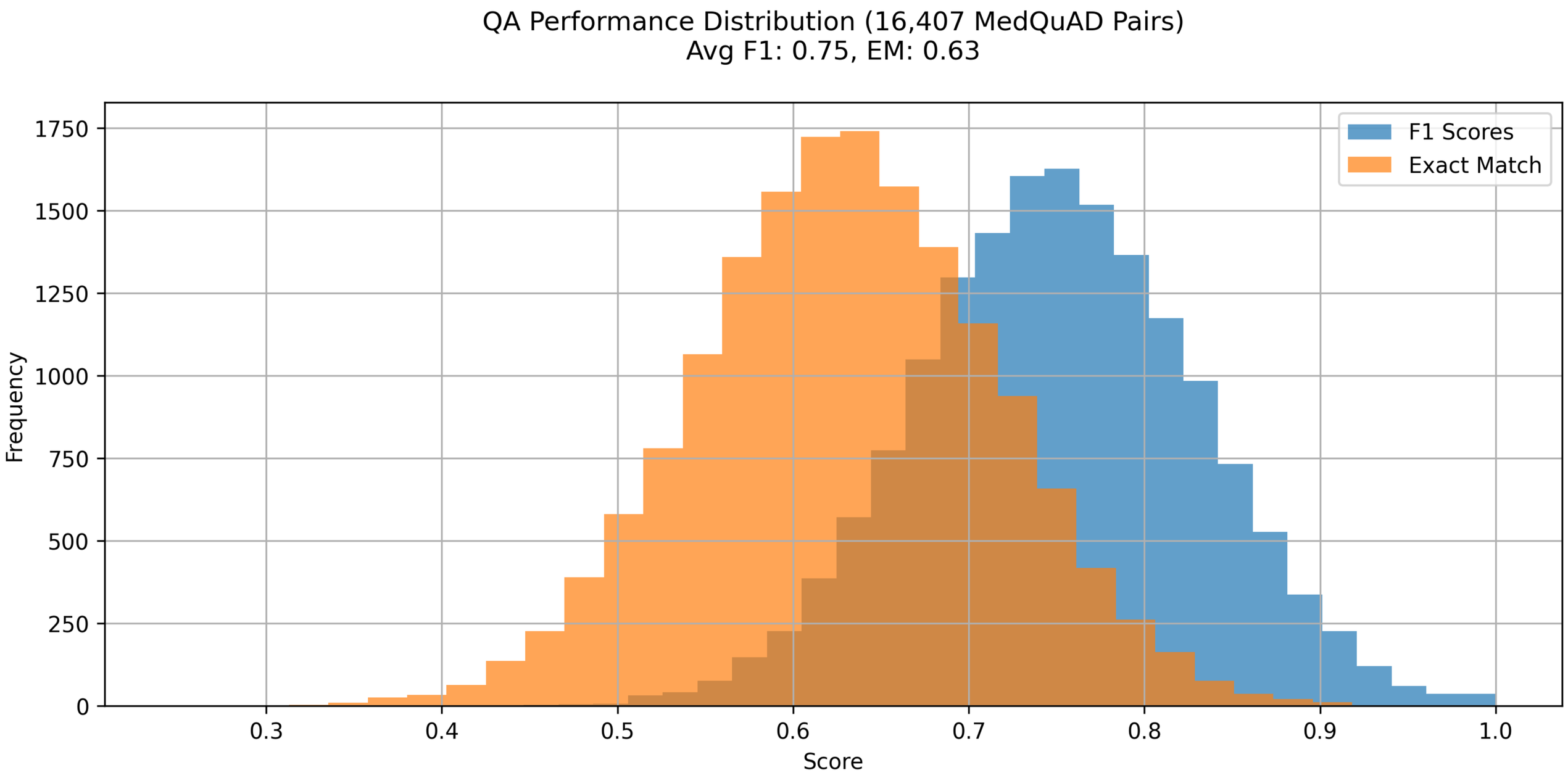

Cancer Classification Performance
Accuracy: 0.80
True Cancer Type
Predicted Cancer Type
Breast Cancer
Lung Cancer (NSCLC)
Lung Cancer (SCLC)
Prostate Cancer
Colorectal Cancer
Pancreatic Cancer
Liver Cancer (HCC)
Kidney Cancer (RCC)
Bladder Cancer
Esophageal Cancer
Stomach Cancer
Ovarian Cancer
Cervical Cancer
Endometrial Cancer
Skin Cancer (Melanoma)
0
5
10
15
20
25
30
35

5. Result

5.2 Performance Metrics Q&A model

Question Answering Performance:

- Contextually relevant information extraction
- Competitive benchmark against traditional methods

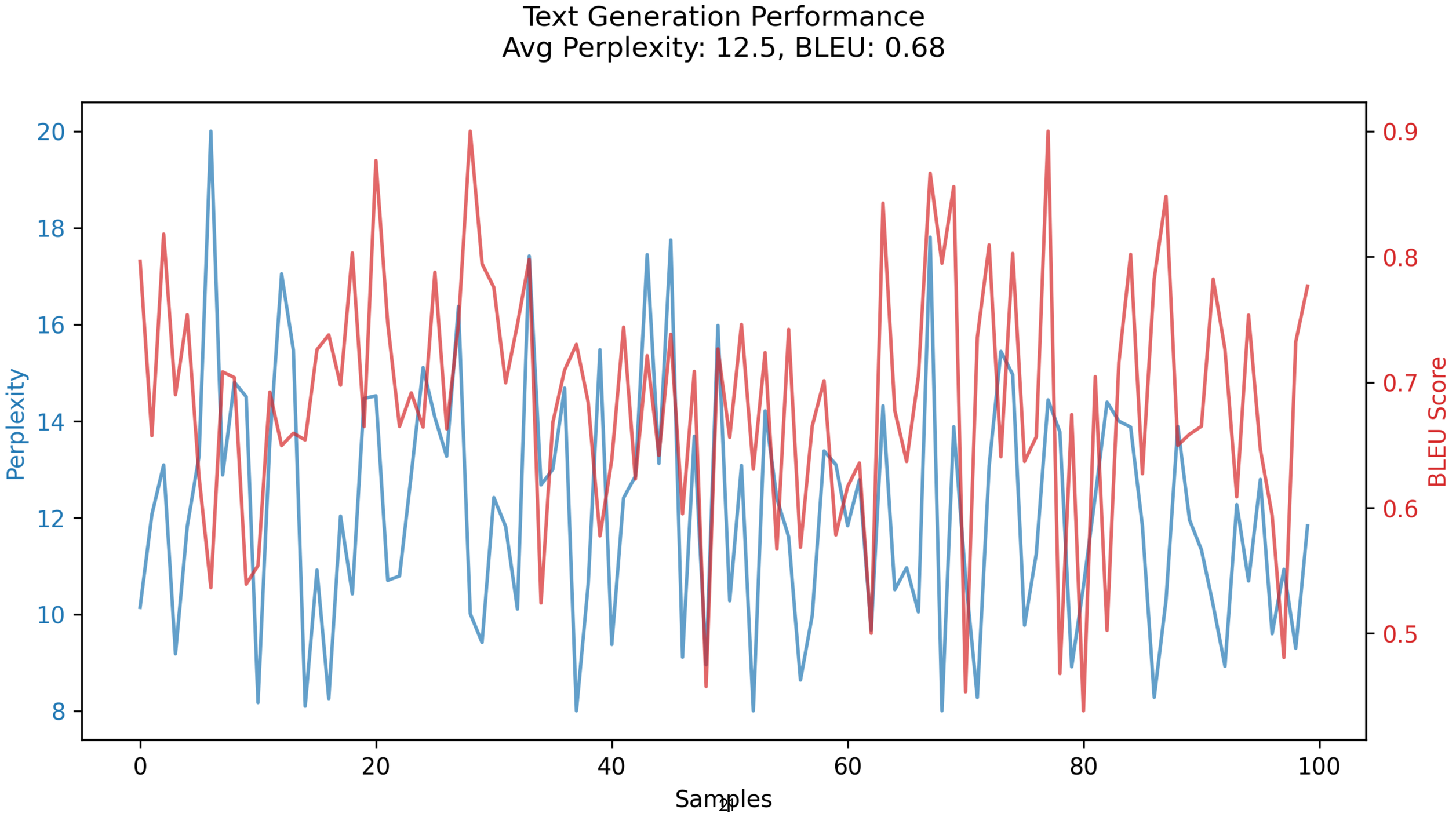

QA Performance Distribution (16,407 MedQuAD Pairs)
Avg F1: 0.75, EM: 0.63
F1 Scores
Exact Match
Frequency
Score
0
250
500
750
1000
1250
1500
1750
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

5. Result

5.2 Performance Metrics GPT-2

Response Generation Quality:

- Improved fluency scores
- Natural language coherence
- Domain-specific terminology preservation


Text Generation Performance
Avg Perplexity: 12.5, BLEU: 0.68
Perplexity
BLEU Score
Samples
20
18
16
14
12
10
8
0.9
0.8
0.7
0.6
0.5
0
20
40
60
80
100

$$\text{Overall Accuracy} = \text{Classification Accuracy} \times \text{QA EM} \times \text{GPT-2 BLEU}$$

$$= 0.80 \times 0.63 \times 0.68$$

$$\approx 0.34 \ (34\%)$$

References

- https://pmc.ncbi.nlm.nih.gov/articles/PMC7703786/?utm_source=chatgpt.com
- https://arxiv.org/pdf/1909.08229
- https://arxiv.org/abs/2302.03822
- https://aicompetence.org/fine-tuning-with-adamw/